Write a Trait Quality-Control Report

Description

Computes per-domain and per-domain×taxon summary statistics for raw trait data and identifies suspected outlier taxa at two levels: a lenient domain-level check and a stricter within-taxon check. Writes the per-taxon summary as a human-readable CSV report and creates a header-only corrections template when that template does not yet exist.

Usage

```
write_trait_quality_control_report(

  data_trait_records,

  path_trait_corrections = here::here("Data/Input/trait_manual_corrections.csv"),

  path_trait_quality_control_report = NULL,

  domain_iqr_multiplier = 3,

  taxon_iqr_multiplier = 1.5,

  minimum_taxon_records = 10L

)
```

Arguments

`data_trait_records`	A data frame in long format with at least the columns 'taxon_name', 'trait_domain_name', 'trait_name', and 'trait_value'.
`path_trait_corrections`	Character scalar. Path to 'trait_manual_corrections.csv'. Default: 'here::here("Data/Input/trait_manual_corrections.csv")'.
`path_trait_quality_control_report`	Character scalar or 'NULL'. Optional path to the CSV quality-control report. If 'NULL' (default), the function writes a date-stamped file to 'Data/Temp/trait_quality_control_report_YYYY-MM-DD.csv'.
`domain_iqr_multiplier`	Positive numeric scalar. A domain value is flagged as a suspected outlier when '|trait_value - median| > domain_iqr_multiplier * IQR', computed across all records in that domain regardless of taxon. Default: '3'. This conservative symmetric median-IQR screen is intended to catch unit mix-ups while tolerating broad cross-taxon variation.

`taxon_iqr_multiplier`	Positive numeric scalar. The stricter IQR multiplier applied within each taxon × domain group. A record is flagged when '|trait_value - taxon_median| > taxon_iqr_multiplier * taxon_IQR' and the taxon has at least 'minimum_taxon_records' records in the domain. Default: '1.5'. This is tighter than the domain-level 'domain_iqr_multiplier' because within-taxon variability should be far smaller than cross-taxon variability; any record deviating by more than 1.5 × the within-taxon IQR is suspicious even when it is not an extreme cross-domain outlier.
`minimum_taxon_records`	Positive integer scalar. Minimum number of records a taxon must have within a domain before the per-taxon IQR check is applied. Taxa with fewer records are skipped at the taxon level (IQR cannot be estimated reliably from very small samples). Default: '10L'.

Details

Two outlier detection levels are applied:

1. **Domain level** ('suspected_outlier_taxa_domain'): flags values where '|trait_value - domain_median| > domain_iqr_multiplier * domain_IQR' (default 3). Applied to all records.

2. **Taxon level** ('suspected_outlier_taxa_taxon'): flags values where '|trait_value - taxon_median| > taxon_iqr_multiplier * taxon_IQR' (default 1.5). Applied only when the taxon has at least 'minimum_taxon_records' records in the domain and when 'taxon_IQR > 0'. This stricter check catches within-taxon inconsistencies that the cross-taxon check would miss.

The CSV report contains the per-domain×taxon summary. By default it is written to 'Data/Temp/trait_quality_control_report_YYYY-MM-DD.csv', but an explicit 'path_trait_quality_control_report' can be supplied when the caller needs an isolated output location. The corrections template written when absent contains the columns: 'taxon_name', 'trait_domain_name', 'action', 'scale_factor', 'notes', 'CHECKED'.

Value

A named list with four computed quality-control elements:

'data_summary_by_domain'

A tibble with one row per 'trait_domain_name' containing: 'n_records', 'n_taxa', 'mean', 'median', 'sd', 'quantile_05', 'quantile_95', 'iqr', 'n_suspected_outliers'.

'data_summary_by_domain_taxon'

A tibble with one row per 'trait_domain_name × taxon_name' combination where the taxon has at least 'minimum_taxon_records' records in that domain, containing: 'n_records', 'mean', 'median', 'sd', 'iqr', 'n_suspected_outliers'.

'vec_suspected_outlier_taxa_domain'

A character vector of taxon names whose trait value in any domain falls more than 'domain_iqr_multiplier' × IQR from the domain median (cross-taxon check).

'vec_suspected_outlier_taxa_taxon'

A character vector of taxon names that have at least 'minimum_taxon_records' records in a domain and whose trait value falls more than 'taxon_iqr_multiplier' × IQR from their own taxon median (within-taxon check).

See Also

[flag_trait_outliers()], [load_trait_corrections()], [validate_trait_corrections()], [correct_trait_records()]

Examples

```
data_trait_records <-

  tibble::tibble(

    taxon_name = base::rep("Quercus", 12L),

    trait_domain_name = base::rep("SLA", 12L),

    trait_name = base::rep("LMA", 12L),

    trait_value = base::seq(10, 21, by = 1)

  )


path_trait_corrections <-

  base::tempfile(fileext = ".csv")


path_trait_quality_control_report <-

  base::tempfile(fileext = ".csv")


write_trait_quality_control_report(

  data_trait_records = data_trait_records,

  path_trait_corrections = path_trait_corrections,

  path_trait_quality_control_report = path_trait_quality_control_report

)


write_trait_quality_control_report(

  data_trait_records = data_trait_records,

  path_trait_corrections = path_trait_corrections
```

)